

Bhuvan Veeravalli

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Summary

CS student at UAlberta (graduating May 2027) with 1.5+ years of NSERC-funded ML research experience building production-grade deep learning systems for thermal forecasting and document intelligence. Specialized in transformer architectures, LLM orchestration, and end-to-end model deployment.

Education

BSc Computing Science Specialization | GPA: 3.8/4.0

Expected May 2027

University of Alberta

Edmonton, AB

- **Relevant Courses:** Machine Learning I & II, Reinforcement Learning, Search & Planning in AI, Natural Language Processing, Algorithms I, Applied Statistics I & II, Linear Algebra I & II.

Work Experience

FES Summer Intern

Apr 2026 – Present

Future Energy Systems

Edmonton, AB

- Developing **spatiotemporal deep learning models** to predict 3D thermal dynamics in molten salt energy storage systems across 30+ experimental configurations.
- Processing **10,000+ infrared image frames** to extract thermal gradient patterns and heat-transfer behavior, feeding structured training pipelines for thermal forecasting models.
- Investigating correlations between multi-sensor measurements and infrared image features to improve predictive accuracy of dynamic thermal response models.

ML Research Assistant

Jan 2025 – Present

Renewable Thermal Energy Lab

Edmonton, AB

- Developed **multi-head attention LSTM** with transformer-style architecture to predict 10-point thermal storage temperature profiles, achieving $R^2 = 0.98$ and **MAE** $< 0.5^\circ\text{C}$ across 150+ experimental configurations.
- Engineered automated **data ingestion pipeline** integrating real-time sensor streams with residual analysis workflows, reducing model validation time by **40%**.
- Implemented **adaptive error-tracking** system to identify optimal test configurations, improving diagnostic precision and accelerating identification of performance thresholds.

Projects

LAIrner — Socratic AI Tutoring Platform | *Lead AI Systems Engineer*

May 2026 – Jun 2026

- Designed a **dual-model LLM orchestration pipeline** using **GPT-4o** for Socratic reasoning and a zero-shot **GPT-4o-mini** classifier for real-time subject categorization across 14 domains with sub-200ms latency.
- Engineered **prompt guardrails with negative constraints** to prevent answer leakage; adaptive feedback loop adjusts hint density per attempt count — an early-stage **RLHF-inspired alignment** mechanism.
- Built stateful backend with **Next.js 16**, **Prisma**, and **Turso** (libSQL) to atomically ingest interaction metrics, chat histories, and proficiency scores in a single database transaction.

IntelliDocs — Document Intelligence System | *Lead AI Engineer*

Jan 2025 – Apr 2025

- Architected a **RAG system** using **Mistral-7B** with hybrid retrieval (BGE dense embeddings + BM25), fine-tuned via **QLoRA** on 2,000+ synthetic Q&A pairs and human-annotated examples.
- Achieved **0.89 NDCG@10**, improved answer relevance by **35%**, and reduced hallucination rate by **40%** vs. baseline zero-shot Mistral-7B; deployed inference pipeline via **FastAPI** with sub-second query response.

FlightPredict — Delay Prediction System | *ML Systems Engineer*

Sep 2024 – Dec 2024

- Built a **multi-target PyTorch MLP** over 5M+ flight records (BTS + NOAA), deployed via **FastAPI/Spring Boot**; achieved **85% accuracy** with **sub-500ms inference latency**.

Skills

- **AI/ML & GenAI:** PyTorch, TensorFlow, Transformer Architectures, LLMs, RAG, LangChain, NLP, QLoRA Fine-tuning, Prompt Engineering, RLHF, Inference Optimization, Model Evaluation (NDCG, MAE, R^2), Vector Databases
- **Programming:** Python, Java, JavaScript, C++, C, C#
- **Backend & MLOps:** FastAPI, Spring Boot, Node.js, Docker, Kubernetes, CI/CD, AWS (Lambda, S3, Bedrock), GCP, PostgreSQL, MongoDB, Git, Linux
- **Frontend:** ReactJS, Next.js, Tailwind CSS

Awards & Recognition

- **URI Stipend Award — Future Energy Systems** Apr 2026
- **NSERC Undergraduate Student Research Award** Apr 2025
- **Dean's Honor Roll** Jun 2024, Jun 2025
- **Generative AI for Software Development — DeepLearning.AI** Oct 2025
- **Google Cloud Data Analytics Certificate — Google** Sep 2025